

BUBBLE BLASTER

Controlling the submarine

The next stage of the program is to write the code that makes the submarine move when the arrow keys are pressed. The code will create a function called an “event handler”. The event handler checks which key has been pressed and moves the submarine.

3 Type this code to create a function called “move_ship”. This function will move the submarine in the correct direction when a cursor key is pressed. Try running it to see how it works.

```
SHIP_SPD = 10  # The sub will move this far when a key is pressed
def move_ship(event):
    if event.keysym == 'Up':
        c.move(ship_id, 0, -SHIP_SPD)
        c.move(ship_id2, 0, -SHIP_SPD)
    elif event.keysym == 'Down':
        c.move(ship_id, 0, SHIP_SPD)
        c.move(ship_id2, 0, SHIP_SPD)
    elif event.keysym == 'Left':
        c.move(ship_id, -SHIP_SPD, 0)
        c.move(ship_id2, -SHIP_SPD, 0)
    elif event.keysym == 'Right':
        c.move(ship_id, SHIP_SPD, 0)
        c.move(ship_id2, SHIP_SPD, 0)
c.bind_all('<Key>', move_ship)
```

Moves the two parts of the sub up when the up arrow key is pressed

These lines are activated when the down arrow key is pressed, and the sub moves down

The sub moves left when the left arrow key is pressed

Moves the sub right when the right arrow key is pressed

Tells Python to run “move_ship” whenever any key is pressed

y coordinate gets smaller moving up



Don't forget to save your work

► How it works

The “move_ship” function moves the sub in different directions. Adding to the sub's x and y coordinates moves it right and down, while subtracting from them moves it left and up.

x coordinate gets smaller going left

